Why can we resist smoking now for future health?

From discounted utility theory to preference reversal

People do not value future benefits at their full amount: they discount them over time and convert them into “present value.”

Discounted utility theory

Smoking provides the utility of “feeling good right now.” The health benefits of quitting, however, appear in the future. People discount that future utility over time and compare it with present utility when deciding.
Present value = future utility ÷ (1 + r)^t
r: rate of time preference; t: time until the future benefit.

What is the rate of time preference?

A higher rate puts greater weight on “now rather than later.” Heavily discounting future health makes an immediate cigarette more attractive. With a low rate, future health retains a high present value, making it easier to resist current pleasure.
High rate: prioritize a cigarette now; heavily discount future health.
Low rate: prioritize future health; find it easier to resist present pleasure.

Why does the intention to quit collapse?

As the moment approaches, the valuation of utility itself changes

Preference reversal

Today you may think, “I will quit next month. Health matters more.” But next month, with a cigarette right in front of you, this can become “It is fine to smoke just today.” In the distant future, choose “health” → temptation comes close → choose “a cigarette now.”

Time inconsistency

The earlier plan to quit does not match the actual choice when that moment arrives. This loss of consistency in decision-making as time passes is called time inconsistency.

The overall process

Large utility from future health → discount by the rate of time preference → convert to present value and compare with immediate pleasure → as temptation approaches, present pleasure becomes relatively stronger → preference reversal → time inconsistency.
People who can resist now for future health do not discount that future health very heavily.